

Contact

habibapez@hotmail.com

www.linkedin.com/in/habib-apez
(LinkedIn)

github.com/HabibApez (Other)

Top Skills

Python (Programming Language)

Machine Learning

Artificial Intelligence (AI)

Languages

Spanish (Native or Bilingual)

French (Limited Working)

German (Elementary)

English (Professional Working)

Chinese (Elementary)

Certifications

5-Day Automotive Cybersecurity
Bootcamp Certificate

Exploring the Future of Development
with GitHub Copilot Workspace

XACS101 - Foundations of
Information Security

Play It Safe: Manage Security Risks

Honors-Awards

Graduated with honors from Instituto
Politécnico Nacional

Megacities Entrepreneurship Award
2019

Publications

Design and implementation of a
digital control for a quadcopter axis
from the perspective of a seesaw

Habib Apez

Sr ADAS R&D Embedded Software Engineer | AI Enthusiast |
Cybersecurity | AUTOSAR | SAFe SM | C, C++ & Python | Linux
Mexico City, Mexico

Summary

Embedded Software Engineer with 8+ years of experience in ADAS and automotive R&D, specializing in short- and long-range radars, safety-critical systems, AUTOSAR architecture, and cybersecurity. I've spent my career building reliable software that powers the next generation of advanced driver-assistance systems, from requirement decomposition and detailed design to hands-on C/C++ development, static analysis, unit test and ECU-level debugging. Beyond writing code, I bring a dual perspective as a Certified SAFe® 6 Scrum Master, having led Agile ceremonies, coached cross-functional teams to improve delivery and collaboration, and collaborated with global stakeholders across the full V-Cycle development process. My technical foundation spans MISRA C, ISO 26262, ISO 21434, UDS, CAN/Ethernet protocols, cryptography, and tools like Lauterbach, CANoe, DaVinci Configurator, and Elektrotbit. I hold certifications in Automotive Cybersecurity (Cyber4People), Information Security (Stanford Online), Cybersecurity (Google), and AI-assisted development with GitHub Copilot (Microsoft). Actively shaping AI adoption in the automotive industry: I serve as a VIO Champion driving AI tool adoption across AUMOVIO, participate in the AI R&D MX Committee, contribute to Project Expedition 26 (a hands-on initiative building AI agents with LangChain) and am co-organizing and coaching at the UX Promptathon 4.0. Electronics & Communications Engineer with specialization in Control, graduated with honors from Instituto Politécnico Nacional. Completed multiple competitive research internships across Mexican universities (IPN, Universidad de Colima, Instituto Tecnológico de Celaya) and internationally at McMaster University, Canada, where I developed signal acquisition and data processing systems for UAVs. C2 English (EF SET 84/100), with working knowledge of French, elementary knowledge of German and currently learning Chinese. Co-founded Aparcar MX, "The Airbnb for parking slots", a Continental-backed startup recognized with the Megacities Entrepreneurship Award

2019. 5+ years of volunteer work in environmental sustainability and social responsibility across organizations including Continental, Grupo Modelo, and Guerreros Guardianes. My next step: pursuing a Master's in Artificial Intelligence with a research focus in Canada, bridging my deep automotive embedded systems expertise with AI to push the boundaries of intelligent, safety-critical systems.

Experience

AUMOVIO

Sr ADAS R&D Embedded Software Engineer

September 2025 - Present (9 months)

Querétaro, Mexico

Continental

7 years 8 months

Sr. ADAS R&D Embedded Software Engineer

November 2021 - September 2025 (3 years 11 months)

Querétaro, Mexico

Responsibilities:

- * Analyze L3 requirements and decompose them into L4 requirements.
- * Create and maintain detailed designs for SW components.
- * Design and execute test cases based on requirements.
- * Develop hand-written source code following MISRA C and ADAS coding guidelines.
- * Configure BSW modules in Tresos and SWCs in DaVinci Configurator.
- * Create and maintain SWCs in AUTOSAR builder.
- * Perform static code analysis using QAC and unit testing with Courage or Google Test, generating coverage reports.
- * Conduct peer code reviews and bench testing using debuggers (white box testing) and CANoe restbus (black box testing).
- * Perform smoke tests on the ECU after changes are implemented.
- * Submit code changes to CI platforms (Jenkins, GitHub, IMS).
- * Track issues, document progress, and findings in JIRA or IMS tickets.
- * Identify root-cause problems through troubleshooting and ECU debugging.
- * Investigate, debug, document, and fix SW issues.
- * Provide technical guidance to other team members.
- * Participate in delta code reviews, configuration reviews, and design reviews.
- * Collaborate with other SW Development, SW Testing, and Systems teams.

- * Maintain constant communication with the Product Owner and Scrum Master.
- * Collaboration with global teams on various projects.

Other knowledge:

AUTOSAR, DEM, NVM, Cybersecurity, APAR, FCU, COM, RTE, OS, MICROSAR, Real-time constraints, SUM Standard Utility Modules, arxml files, Software Development Life Cycle, V-Cycle Software Development process, Unified Diagnostic Services (UDS), ISO 14229, ISO 26262, ISO 21434, ASPICE, SAFe, Cryptography, CAN and Ethernet protocols, Python, ARXML.

Tools:

Visual Studio Code, Github Copilot, TD5, Visual Studio, DOORS, MKS, IMS, Jenkins, Lauterbach debugger, CANoe, AUTOSAR Builder, AUTOSAR Explorer, DaVinci Configurator, Vector Tools, GitHub, Git, Git Extensions, QAC, Courage, Google Test, Google Mocks, Plant UML, Doxygen, Tresos EB, JIRA, Beyond Compare.

SW Scrum Master

May 2021 - November 2023 (2 years 7 months)

Querétaro, Mexico

Responsibilities:

- * Coordinate sprint planning, retrospective meetings, and daily stand-ups to ensure effective team collaboration.
- * Facilitate internal communication and served as the primary point of contact for external stakeholders.
- * Work closely with the Product Owner to manage the Sprint Backlog and address new requests.
- * Resolve conflicts and removed obstacles to help the team implement changes effectively.
- * Ensure deliverables met quality standards at the end of each sprint.
- * Guide the development team to higher Scrum maturity, fostering a collaborative and productive team environment.
- * Report on team progress, including generating velocity charts and burn-down reports.
- * Manage the team's Kanban board in JIRA and tracked Agile metrics.

Tools:

JIRA, Confluence, EasyBI

Full ADAS R&D Embedded Software Engineer

October 2019 - November 2021 (2 years 2 months)

Querétaro, Mexico

Intermediate ADAS R&D Embedded Software Engineer

February 2018 - October 2019 (1 year 9 months)

Querétaro, Mexico

Aparcar MX

Co-Founder and CSO

July 2018 - February 2019 (8 months)

Mexico City, Mexico

"The Airbnb for parking slots" in megacities.

With Fernando Paulin and Carolina Acosta.

A startup powered by Continental

co-pace GmbH

Entrepreneur at Co-Pace

September 2018 - January 2019 (5 months)

Mexico City, Mexico

Continental

Embedded Software Engineering Trainee

October 2017 - December 2017 (3 months)

Querétaro, Mexico

Automotive Entry Program.

Responsibilities:

- * Develop a Power Window Lifter Control Unit and an Airbag Control Unit demo project on a 32-bit microcontroller (S32K144).
- * Follow SDLC and V-Cycle methodologies.
- * Analyze and decompose requirements.
- * Create and conduct test cases based on the requirements (white box and black box testing).
- * Develop the code in C language based on the requirements with a layer programming mindset.
- * Have periodical reviews with stakeholders (Continental engineers).
- * Keep track of the changes using version control in GitHub.
- * Implement a Binary Progression Algorithm scheduler using RTOS concepts.
- * Integrate third-party communication drivers for LIN and CAN into the project.
- * Use state machines and interrupt handling.
- * Follow MISRA C and naming convention.
- * Code low-level drivers for uC peripherals.
- * Debugging and troubleshooting real time embedded systems.

Tools:

IAR, S32 Studio, Git, debugger, logic signal analyzer, CAN analyzer, oscilloscope and signal generator

Siemens

Engineering Trainee

February 2016 - August 2016 (7 months)

Mexico City, Mexico

Business Learning Program.

Responsibilities:

- * Provide support to different Business Units of the company, mainly to the strategy's division.
- * Statical determination of the FTWC for Mesoamerica's region.

Instituto Tecnológico de Celaya

Research Intern

June 2015 - August 2015 (3 months)

Celaya, Guanajuato, Mexico

Department of Mechanics Engineering.

Bluetooth monitoring system for a sensor network on a hybrid vehicle using dsPIC30F4013 with C30 and LabVIEW.

Professor: PhD Francisco Javier Pérez Pinal.

Instituto Politécnico Nacional

Electronics Laboratory Technician Intern

August 2014 - January 2015 (6 months)

Mexico City, Mexico

Responsibilities:

- * Support scientists, teachers and students to carry out electronic tests.
- * Arrange laboratory equipment.
- * Inventorying.

McMaster University

Research Intern

June 2014 - August 2014 (3 months)

Hamilton, Ontario, Canada

Department of Electrical and Computer Engineering.

Monitoring, acquisition and data processing for a quadcopter using UMI-MPU6050 and dsPIC30F4013 with C30 and LabVIEW.

Professor: PhD Francisco Javier Pérez Pinal.

Universidad de Colima

Research Intern

June 2013 - August 2013 (3 months)

Coquimatlán, Colima, Mexico

Faculty of Mechanics and Electronics Engineering.

Digital temperature control system (PID) embedded in a 18-bit microcontroller.

Professor: MC and MTI Andrés Gerardo Fuentes Covarrubias.

Universidad de Colima

Research Intern

June 2012 - August 2012 (3 months)

El Naranjo, Colima, México

Faculty of Mechanics and Electronics Engineering.

Trajectory and patterns' design for cooperative and coordinated industrial KUKA robots.

Professor: Dr. Jorge Gudiño Lau.

Education

Instituto Politécnico Nacional

Bachelor's degree, Electronics and Communications Engineering with specialization in Control · (2010 - 2015)

Universidad Marista de Guadalajara

Certificate, Embedded Linux · (November 2022 - February 2023)

ITESO Universidad Jesuita de Guadalajara

Certificate, Security for Embedded Automotive Systems · (January 2022 - August 2022)

Tecnológico de Monterrey

Certificate, TEC LEAN Launch, Incubator Program · (2018 - 2018)

Universidad Nacional Autónoma de México

Certificate, Embedded Systems · (2017 - 2017)